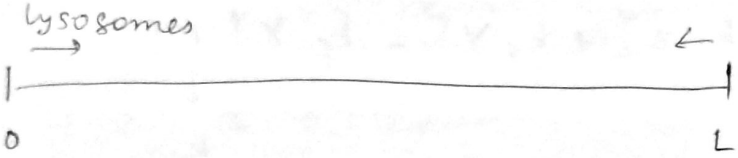
Lysosome-phagosome interaction model with reversals



Lysosomes produced at $x=0$ with production rate k_{PY}

Lysosome velocity: v_Y, reversal rate: k_s

Phagosomes produced at $x = L$ with production rate k_{PP}

Phagosome velocity: v_P, no reversals

EOM:

$$\frac{\partial Y^+}{\partial t} = -v_Y \frac{\partial Y^+}{\partial x} - k_s Y^+ + k_s Y^- - k_f (Pl) Y^+ \quad \text{— (1)}$$

($-v_Y \frac{\partial Y^+}{\partial x}$: lyso moving forward; k_f: fusion rate; $k_s Y^-$: lyso moving backward; (Pl): average number of phagosomes within interaction length)

$$\frac{\partial Y^-}{\partial t} = v_Y \frac{\partial Y^-}{\partial x} - k_s Y^- + k_s Y^+ - k_f Y^- (Pl) \quad \text{— (2)}$$

Boundary conditions:

$$v\, Y^+(x=0) = k_{PY} \quad \text{— (3)} \quad \text{(lysosome production)}$$

$$Y^-(x=L) = 0 \quad \text{— (4)} \quad \text{(No backward moving lyso at domain edge)}$$

Phagosome equations:

$$\frac{\partial P}{\partial t} = v_P \frac{\partial P}{\partial x} \quad \text{— (5)}$$

$$v_P\, P(x=L) = k_{PP} \quad \text{— (6)} \quad \text{(phagosome production)}$$

at steady state:

$$\frac{\partial P}{\partial t} = 0 \Rightarrow \frac{dP}{dx} = 0 \Rightarrow P(x) = \text{constant}$$

from b.c. $\boxed{P(x) = \frac{k_{PP}}{v_P}}$ — (7)

steady state solution for phago lysosomes:

$$\frac{\partial Y^+}{\partial t} = 0 \Rightarrow -v_y \frac{dY^+}{dx} - k_s Y^+ + k_s Y^- - k_f Y^+ \rho l = 0$$

$\rho l \rightarrow \frac{k_{pp}}{v_p}$

$$\boxed{\frac{dY^+}{dx} = -\left(\frac{k_s + k_f k_{pp} l / v_p}{v_y}\right) Y^+ + \frac{k_s}{v_y} Y^-} \quad — (8)$$

$$\frac{\partial Y^-}{\partial t} = 0 \Rightarrow v_y \frac{dY^-}{dx} - k_s Y^- + k_s Y^+ - k_f Y^- \rho l = 0$$

$$\boxed{\frac{dY^-}{dx} = -\frac{k_s}{v_y} Y^+ + \left(\frac{k_s + k_f k_{pp} l / v_p}{v_y}\right) Y^-} \quad — (9)$$

let $\frac{k_s}{v_y} = a$; $\frac{k_f k_{pp} l}{v_p v_y} = b$;

$$\boxed{\frac{d\bar{Y}}{dx} = \underset{\sim}{A}\, \bar{Y}} \quad — (10)$$

where $\bar{Y} = \begin{pmatrix} Y^+(x) \\ Y^-(x) \end{pmatrix}$; $\underset{\sim}{A} = \begin{pmatrix} -(a+b) & a \\ -a & a+b \end{pmatrix}$

let $a+b \equiv c$,

$$\begin{pmatrix} Y^+(x) \\ Y^-(x) \end{pmatrix} = \left(\frac{a k_{py} \sinh\left(\sqrt{c^2-a^2}(L-x)\right)}{\sqrt{c^2-a^2}\cosh\left(\sqrt{c^2-a^2}\,L\right) + c \sinh\left(\sqrt{c^2-a^2}\,L\right)} \right.$$

let $\lambda_1 \equiv a+b = \frac{k_s}{\nu_y} + \frac{k_f k_{pp} l}{\nu_p \nu_y}$

$\lambda_2 \equiv a = \frac{k_s}{\nu_y}$

Solving eq (10) (work 20200724-saurabh.nb)

$$Y^-(x) = \frac{k_{pY}}{\nu_y}\left[\frac{\lambda_2 \sinh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,(L-x)\right)}{\sqrt{\lambda_1^2-\lambda_2^2}\cosh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,L\right)+\lambda_1 \sinh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,L\right)}\right]$$

$$Y^+(x) = \frac{k_{pY}}{\nu_y}\left[\frac{(\lambda_1^2-\lambda_2^2)\cosh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,(L-x)\right)+\lambda_1\sqrt{\lambda_1^2-\lambda_2^2}\sinh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,(L-x)\right)}{(\lambda_1^2-\lambda_2^2)\cosh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,L\right)+\lambda_1\sqrt{\lambda_1^2-\lambda_2^2}\sinh\left(\sqrt{\lambda_1^2-\lambda_2^2}\,L\right)}\right]$$

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some limiting cases:

(1) $k_s = 0$ (no switching)

$\lambda_1 = \frac{k_f k_{pp} l}{\nu_p \nu_y}$; $\lambda_2 = 0$

$\Rightarrow$ $Y^-(x) = 0$

$$Y^+(x) = \frac{k_{pY}}{\nu_y}\left[e^{-\lambda_1 x}\right] \checkmark$$ (matches previous expression)

(2) $\lambda_1 \to \lambda_2$ (low fusion / low phagosome production)

$$Y^-(x) = \frac{k_{pY}}{\nu_y}\left[\frac{\lambda_2 (L-x)}{1+\lambda_2 L}\right] ; \quad Y^+(x) = \frac{k_{pY}}{\nu_y}\left[\frac{1+\lambda_2 (L-x)}{1+\lambda_2 L}\right]$$